



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Fall 2023

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1.	Read the following code and detect different types of errors. <pre>1. include<math.h> 2. include<stdio.h> 3. double Black (float xx, void y, char z){ 4. float k = 0; 5. double p, q=7,8; 6. if(p/2.5 = 2) 7. k++; 8. q = pow(y,2,3); 9. printf("%lf", p); 10. If(p > 2) 11. print("Hi!! I am available"); 12. else 13. printf("Not available : %ld", &p); 14. } 15. double z = black(2.5, 3, hello); 16. return 2; 17. }</pre> <pre>18. void main(){ 19. char a1 = "0"; 20. int b#1 = 10, 2a; 21. int z[12]; 22. for(k = 0; k<=12; k*=2){ 23. scanf("%ld", z[k]); 24. sum+=zz[k]; 25. printf(%d, sum); 26. print(z, 100); 27. } 28. void printf(int z, int size) 29. Float temp=0; 30. Size = size<0? "blue" : green31.; 31. If(size==0) 32. Print(z, 10) 33. Else(double p = Black(5.3,7,PINK)) 34. printf("%lf", p); 35. }</pre>	12
2.	Construct a Non-recursive Predictive Parsing table for the following grammar & also check whether string "00011111" is accepted or not. The sequence of Moves by Predictive Parser in a table (using stack). $S \rightarrow AS \mid BAC$ $A \rightarrow A1 \mid 0A1 \mid 0B1 \mid B$ $B \rightarrow 0B \mid 0 \mid \epsilon$ $C \rightarrow 1 \mid \epsilon$	[4]

3.	Eliminate immediate left recursion from the following grammar.	[3+3]
	(a) $P \rightarrow PP+RsT \mid (PabRT) \mid PghQr \mid P^{*+}QQ$ $Q \rightarrow Q2Ph \mid QPiq \mid RaQ \mid RcbQQ \mid \epsilon$ $R \rightarrow R(2+T) \mid RciTS \mid aRsTP \mid R \mid \epsilon$ $S \rightarrow SSap \mid SP/T \mid S \mid d/SC \mid abSgt$ $T \rightarrow (RT) \mid Ta \mid bT \mid Tc \mid TTE$ b) $X \rightarrow Xo[4]Y \mid XX*ZI \mid Z+YY \mid \epsilon$ $Y \rightarrow Y+2[3]Zi+ht \mid IY/2f \mid Y2*ab \mid \epsilon$ $Z \rightarrow ZOx \mid oZY \mid Zoo \mid ZYot \mid \epsilon$ $I \rightarrow 1 \mid 2 \mid 3 \mid \epsilon$	
4.	Left Factor the following. (a) Consider each word as a single terminal. $A \rightarrow$ Rabbit Laughs $\mid$ Rabbit Loves Radish $\mid$ Rabits talks $\mid$ Rabbit Loving $\mid$ refrening $\mid$ restrained $\mid$ referee $\mid$ reference $\mid$ Rabits $\mid$ Rabbit Lashes $\mid$ refer $\mid$ Rabbit Lost $\mid$ Rabbit Loving $\mid$ Rabbit Lasting	[3+3]
	(b) Consider all the terminals are of single character/number/symbol. $A \rightarrow (2(2+3)) \mid (2+b) \mid (2(+3(4))) \mid (2-3(5)) \mid (2-3) \mid (2+b(1*2)) \mid (2(+2(1/2))) \mid (a) \mid (2(2-6)) \mid (2+4)$ $B \rightarrow$ Savage $\mid$ Satarotical $\mid$ Sample $\mid$ sanny $\mid$ solemn $\mid$ solo $\mid$ solar $\mid$ Saver $\mid$ Saturn $\mid$ san $\mid$ solder $\mid$ Sa	
5.	For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol.	[4+4+4]
(a)	$A \rightarrow +BCc \mid BDrq \mid CtDr \mid Ah \mid c \mid t$ $B \rightarrow xBk \mid CiD \mid d \mid DiA \mid rAg \mid \epsilon$ $C \rightarrow Bq \mid CcAy \mid fAiBqD \mid cA \mid uq \mid \epsilon$ $D \rightarrow e \mid u \mid Djl \mid DBCm \mid \epsilon$	(c) $X \rightarrow +YZc \mid XoM \mid YqZMi \mid XiM5Y \mid \epsilon$ $Y \rightarrow aMh \mid ZguMx \mid XM/ \mid YX2biZ \mid \epsilon$ $Z \rightarrow Xs \mid Y3M \mid 4XY(a) \mid ZXYo \mid Me \mid \epsilon$ $M \rightarrow 2Xq \mid YZrMt \mid MyXY \mid iZYr \mid \epsilon$
(b)	$P \rightarrow aPRt \mid abPi \mid R \mid QRc \mid xyQ$ $Q \rightarrow ijRQr \mid cPQk \mid a \mid +QR \setminus t \mid \epsilon$ $R \rightarrow pPQc \mid yR2 \mid xR/y \mid lQpR \mid \epsilon$	